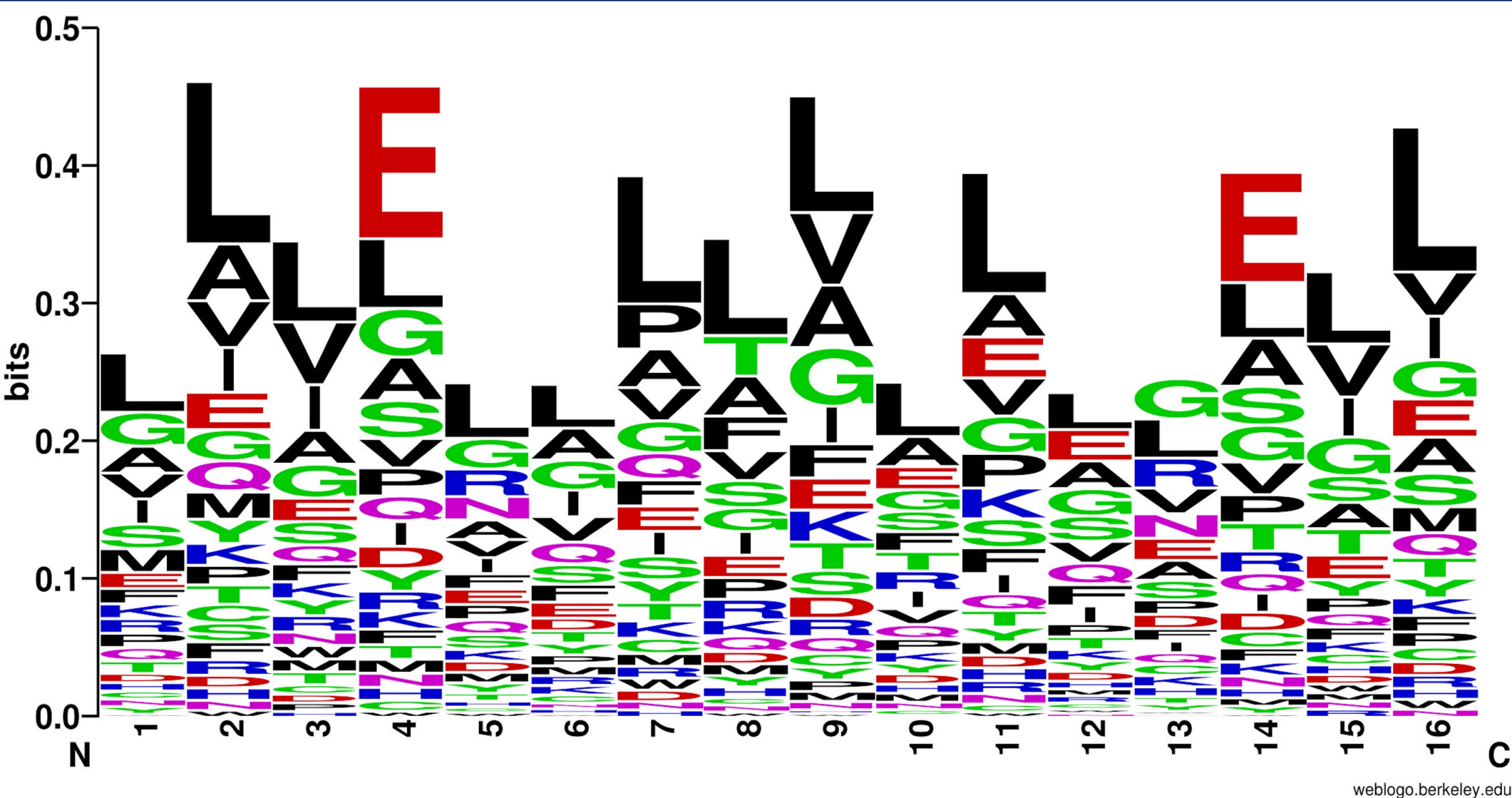


Introduction

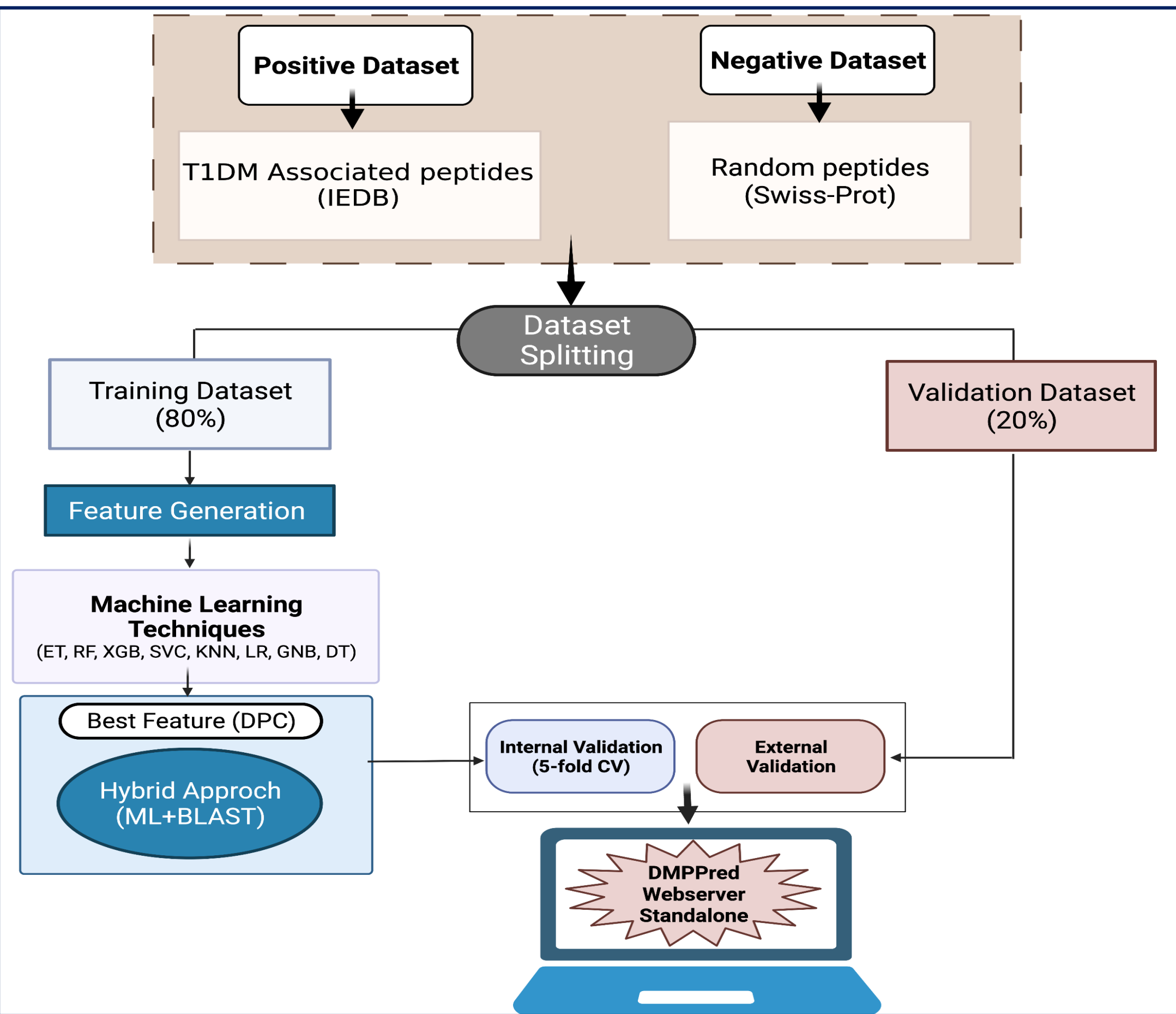
- T1DM is a chronic autoimmune disease where β -cells are destroyed.
- Identifying antigenic peptides that induce T1DM can aid in immunotherapy and vaccine development.
- Limited computational tools to predict T1DM-associated peptides.
- Existing methods (e.g., GPS-MBA) $\rightarrow$ trained on small datasets
- Many antigen-specific immunotherapies in development $\rightarrow$ require accurate identification of risk peptides.
- Compile experimentally validated T1DM-associated peptides.
- Develop ML-based and BLAST-based predictive models.
- A hybrid method using alignment free and alignment based approach
- Build a user-friendly web server for prediction and design

Positional Analysis

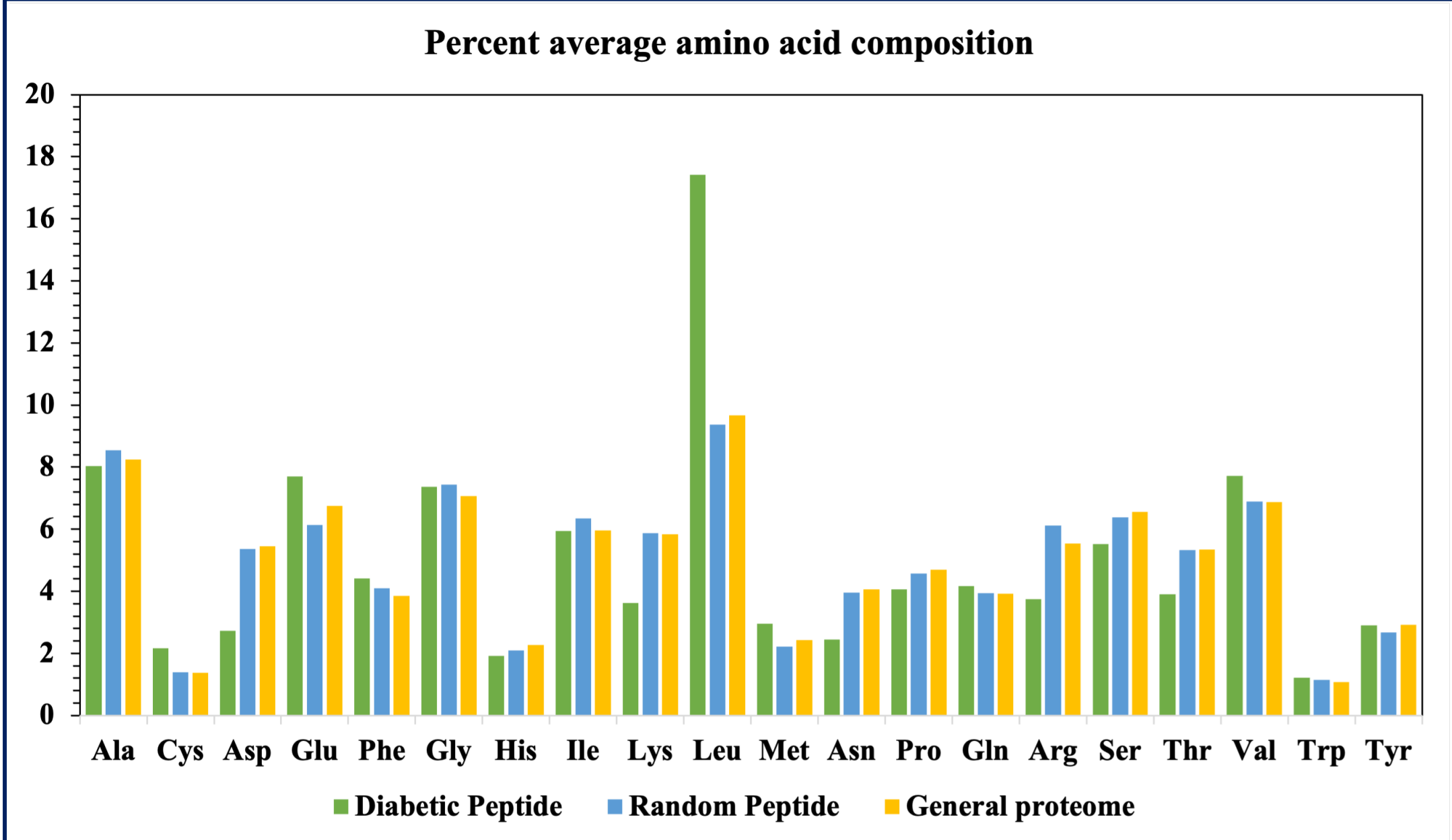


- **Positional Analysis**
 - Leucine is a highly abundant and conserved residue
- **Composition-based analysis**
 - T1DM:- leucine (L), methionine (M), valine (V) and glutamic acid (E) are most abundant amino acid.
- **HLA-binders**
 - HLA-A*02:01 most frequent of HLA alleles in T1DM associated peptides

Methodology



Compositional Analysis



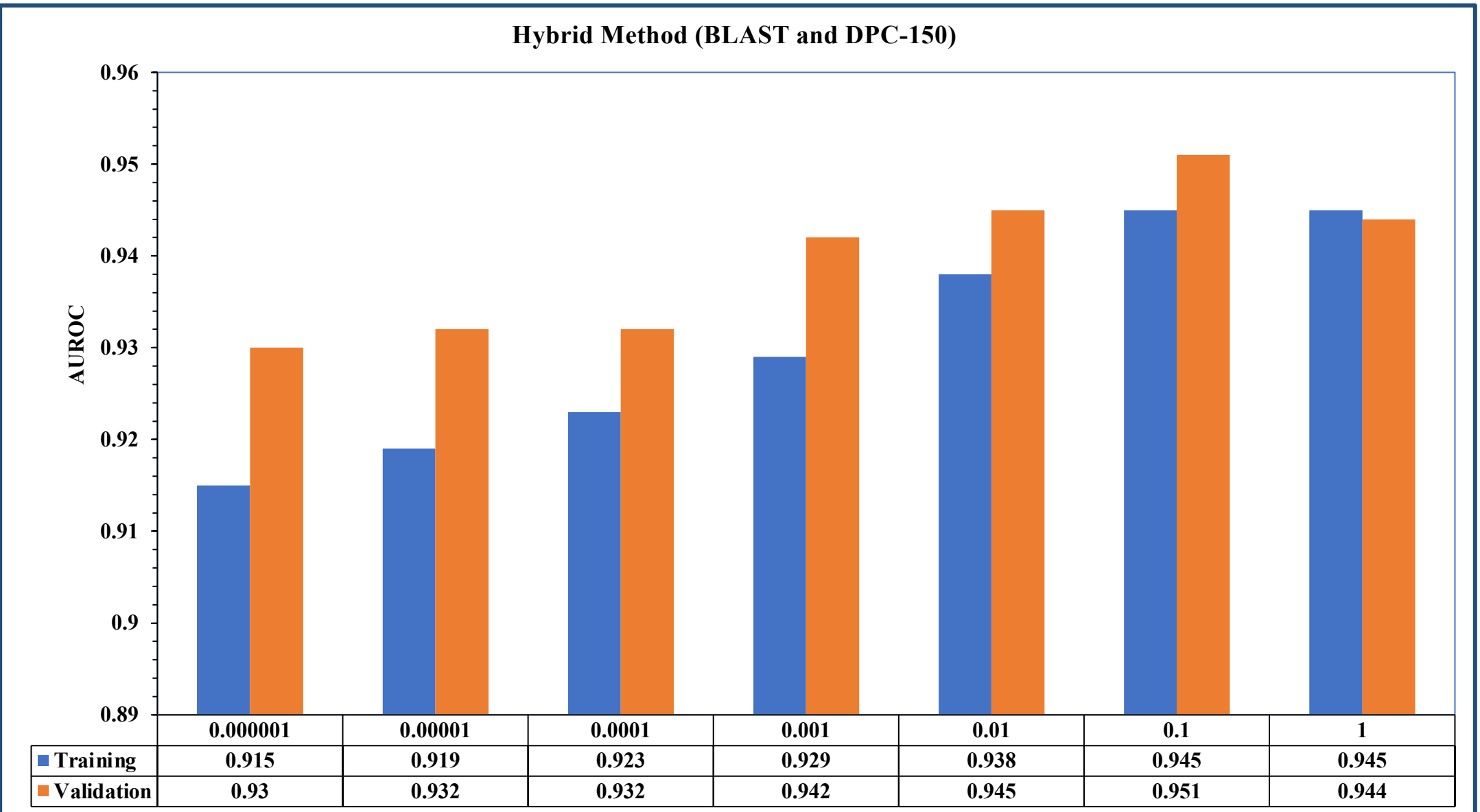
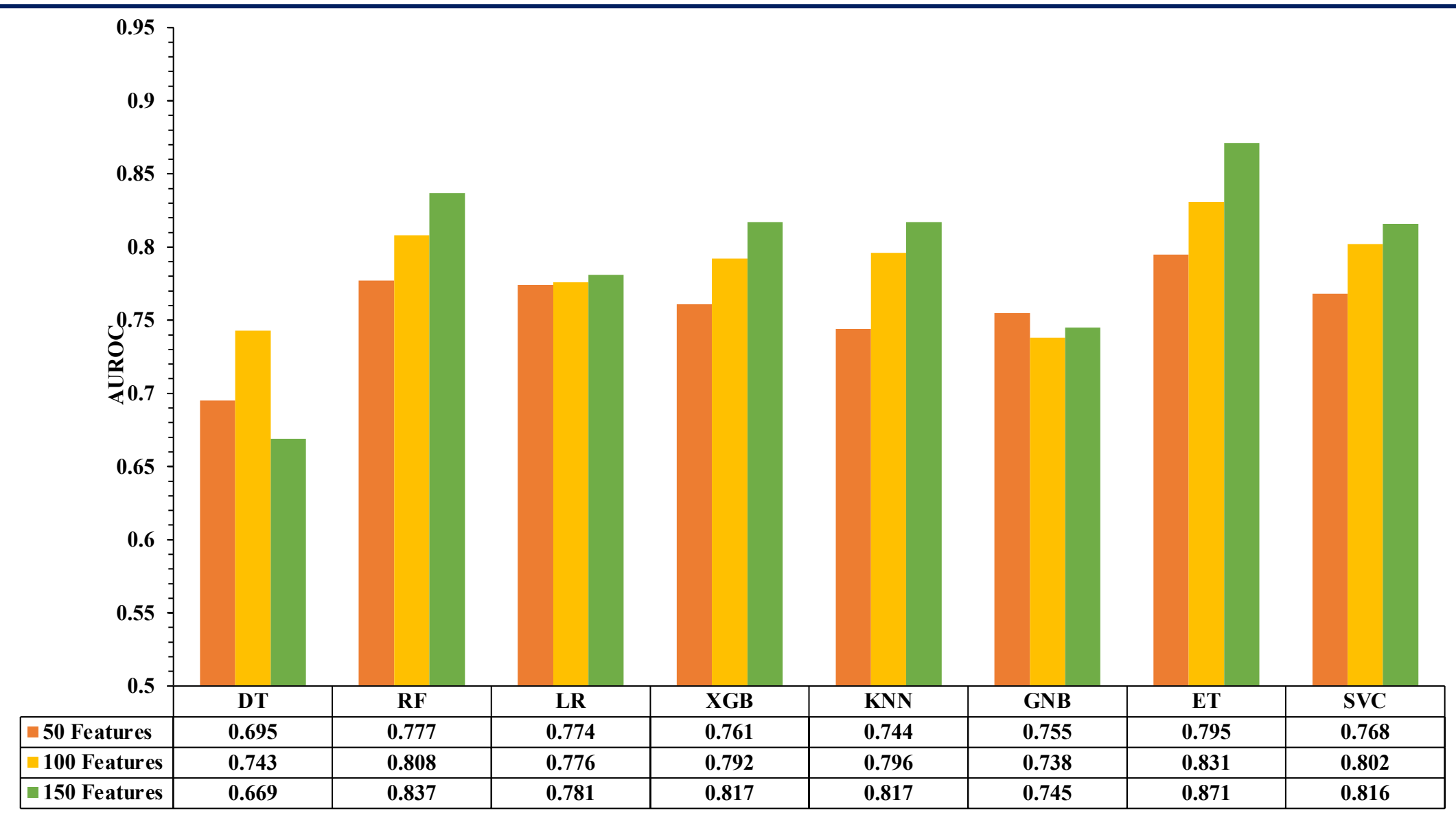
Results

ML Models

- **Best descriptor:** Dipeptide Composition (DPC) -150 Features
- **Best classifier:** Extra Trees (ET)
- Top enriched dipeptides: LL, VE, LV, SL, AL, LQ, LE, IL, ER, LA

Hybrid Model (Combine ML + BLAST)

- Best performance at e-value = 0.1
- Outperformed GPS-MBA (AUROC 0.48)



Conclusion

- DMPPred integrates ML and similarity search for accurate identification of T1DM-associated peptides.
- It supports immunotherapy and vaccine design by identifying potential antigenic regions.
- Access DMPPred at: <https://webs.iitd.edu.in/raghava/dmppred/>